

LGT Toy-Universe Derivation

Physics insert - from light-geometry-time overlap to a minimal H(z) bridge

Purpose

This note turns the light-geometry-time (LGT) kernel into the simplest simulation-ready cosmology. It is not a final proof. It is a controlled toy limit showing how the proposed field overlap can become an effective energy density and then a late-time expansion bridge.

1. Start from the LGT kernel

$$d \tau_{\text{eff}} / d \lambda = 1 + B(\chi) I_{\text{LGT}}$$

$$I_{\text{LGT}} = \eta Q/M_*^2 + \zeta (T_{\text{EM}} \mu_\nu k^\mu k^\nu)/M_*^4 + \gamma (R_{\mu\nu} k^\mu k^\nu)/M_*^2$$

$$Q = \text{grad}^\mu(\theta) \text{grad}_\mu(\omega), \quad k_\mu = \text{grad}_\mu(\theta)$$

Here θ is the eikonal light-flow phase, ω is the RUT phase-order field, Q measures light-flow sampling phase-order, T_{EM} is electromagnetic stress, and $R_{\mu\nu} k^\mu k^\nu$ is curvature projected along the light/null direction.

2. Flat FRW toy universe

$$ds^2 = -dt^2 + a^2(t) d\mathbf{x}_{\text{vec}}^2$$

$$H^2 = \rho_{\text{total}} / (3 M_{\text{Pl}}^2), \quad H = \dot{a}/a$$

Add an LGT sector to the total density:

$$\rho_{\text{total}} = \rho_m + \rho_r + \rho_\Lambda + \rho_{\text{LGT}}$$

$$H_{\text{RUT}}^2 = H_{\text{LCDM}}^2 + \rho_{\text{LGT}}/(3 M_{\text{Pl}}^2)$$

3. Homogeneous light-flow and phase-order

$$\theta = \theta(t), \quad \omega = \omega(t)$$

$$k_\mu = \text{grad}_\mu(\theta) = (\dot{\theta}, 0, 0, 0)$$

$$\text{grad}_\mu(\omega) = (\dot{\omega}, 0, 0, 0)$$

$$Q = \text{grad}^\mu(\theta) \text{grad}_\mu(\omega) = -\dot{\theta} \dot{\omega}$$

$$Q^2 = \dot{\theta}^2 \dot{\omega}^2$$

In this limit, the LGT strength grows when the light-flow rate and phase-order rate are dynamically active together.

4. Mixed interaction and effective density

$$L_{\text{mix}} = -1/2 \xi B(u) Q^2$$

$$L_{\text{mix}} = -1/2 \xi B(u) \dot{\theta}^2 \dot{\omega}^2$$

For a first positive-energy toy sector, define:

$$\rho_{\text{LGT}} = 1/2 \xi B(u) \dot{\theta}^2 \dot{\omega}^2$$

Then the modified Friedmann equation becomes:

$$H_{\text{RUT}}^2 = H_{\text{LCDM}}^2 + [\xi/(6 M_{\text{Pl}}^2)] B(u) \dot{\theta}^2 \dot{\omega}^2$$

5. Bridge activation as a phase transition

$$B(u) = (1+u)/2$$

$$V(u) = \lambda/4 (u^2 - 1)^2$$

$$u(\chi) = \tanh[(\chi - \chi_*)/\Delta\chi]$$

$$B(\chi) = 1/2 \{ 1 + \tanh[(\chi - \chi_*)/\Delta\chi] \}$$

For cosmology, choose the clock variable $\chi = \ln(a) = -\ln(1+z)$. This gives a redshift-space bridge:

$$B(z) = 1/2 \{ 1 + \tanh[(-\ln(1+z) - \chi_*)/\Delta\chi] \}$$

6. Dimensionless expansion law

$$\Omega_{\text{LGT},0} = \xi \cdot \dot{\theta}^2 \cdot \dot{\omega}^2 / (6 M_{\text{Pl}}^2 H_0^2)$$

$$\dot{\theta}^2 \dot{\omega}^2 = \dot{\theta}^2 \dot{\omega}^2 F_{\theta\omega}(z)$$

The simulation-ready toy law is therefore:

$$E_{\text{RUT}}^2(z) = E_{\text{LCDM}}^2(z) + \Omega_{\text{LGT},0} B(z) F_{\theta\omega}(z)$$

Minimal first-pass model:

$$E_{\text{RUT}}^2(z) = E_{\text{LCDM}}^2(z) + \Omega_{\text{LGT},0} B(z)$$

Optional peaked/decaying phase version:

$$F_{\theta\omega}(z) = (1+z)^n \exp(-\beta z)$$

$$E_{\text{RUT}}^2(z) = E_{\text{LCDM}}^2(z) + \Omega_{\text{LGT},0} B(z) (1+z)^n \exp(-\beta z)$$

7. Observable chain and falsifiers

$$H_{\text{RUT}}(z) = H_0 E_{\text{RUT}}(z)$$

$$D_H(z) = c / H_{\text{RUT}}(z)$$

$$D_M(z) = c \int_0^z dz' / H_{\text{RUT}}(z')$$

BAO observables: D_M/r_d , D_H/r_d , D_V/r_d

Requirement	Failure condition
Correspondence limit	If $\Omega_{\text{LGT},0} \rightarrow 0$ or $B(z) \rightarrow 0$ does not recover LCDM.
Parameter economy	If χ_* , $\Delta\chi$, and $\Omega_{\text{LGT},0}$ must be independently re-fit for every probe.
Transfer test	If the same LGT bridge region improves BAO but fails growth/lensing transfer.
Stability	If the effective energy sector creates ghost, gradient, or negative- H^2 pathology.

Reviewer-safe conclusion: the LGT kernel is now derivation-scaffolded into a flat-FRW toy expansion law. It remains provisional until the bridge location, width, amplitude, stability, and multi-probe transfer behavior are constrained.